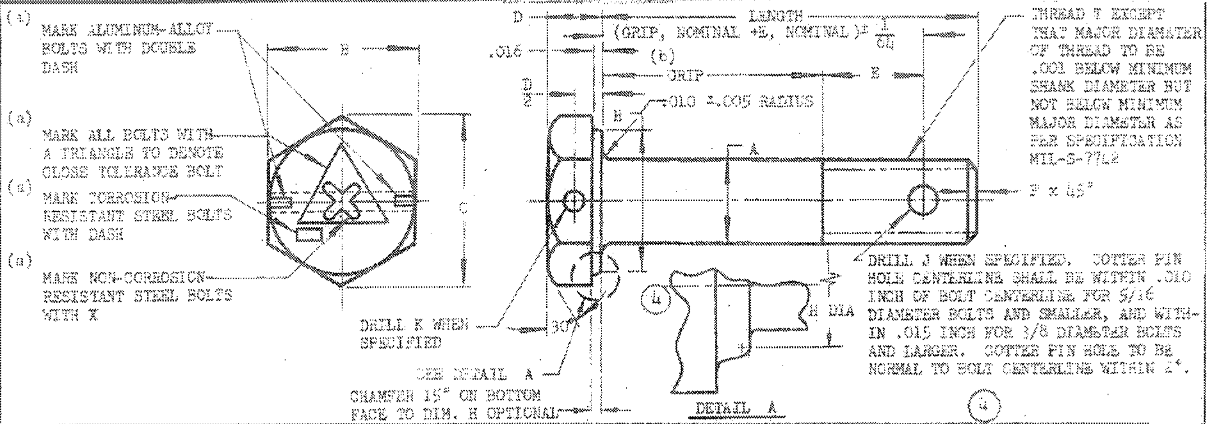


(a) MARK ALUMINUM-ALLOY BOLTS WITH DOUBLE DASH

(a) MARK ALL BOLTS WITH A TRIANGLE TO DENOTE CLOSE TOLERANCE BOLT

(a) MARK CORROSION-RESISTANT STEEL BOLTS WITH DASH

(a) MARK NON-CORROSION-RESISTANT STEEL BOLTS WITH X

DRILL K WHEN SPECIFIED

SEE DETAIL A

CHAMFER 15° ON BOTTOM FACE TO DIM. H OPTIONAL

THREAD T EXCEPT THAT MAJOR DIAMETER OF THREAD TO BE .001 BELOW MINIMUM SHANK DIAMETER BUT NOT BELOW MINIMUM MAJOR DIAMETER AS PER SPECIFICATION MIL-S-7742

DRILL J WHEN SPECIFIED. COTTER PIN HOLE CENTERLINE SHALL BE WITHIN .010 INCH OF BOLT CENTERLINE FOR 5/16 DIAMETER BOLTS AND SMALLER, AND WITHIN .015 INCH FOR 3/8 DIAMETER BOLTS AND LARGER. COTTER PIN HOLE TO BE NORMAL TO BOLT CENTERLINE WITHIN 2°.

④

BASIC AN PART NO.	THREAD T	A DIA		B		C REF	D		E REF	F		(d) H DIA	J DRILL DIA +.010 -.000	K DRILL DIA +.010 -.000
		MAX	MIN	MAX	MIN		MAX	MIN		MAX	MIN	MIN		
AN173	NO. 10-32 NF-3A	.1894	.1889	.377	.365	.430	.141	.109	17/64	.047	.015	.335	.070	.046
AN174	1/4 -28 UNF-3A	.2492	.2487	.440	.428	.510	.172	.140	5/16	.047	.015	.398	.076	.046
AN175	5/16-24 UNF-3A	.3117	.3112	.502	.490	.580	.204	.172	23/64	.063	.031	.460	.076	.070
AN176	3/8 -24 UNF-3A	.3742	.3737	.565	.553	.650	.235	.203	7/16	.063	.031	.523	.106	.070
AN177	7/16-20 UNF-3A	.4367	.4362	.627	.615	.720	.266	.234	31/64	.063	.031	.585	.106	.070
AN178	1/2 -20 UNF-3A	.4991	.4986	.752	.740	.870	.297	.265	39/64	.063	.031	.710	.106	.070
AN179	9/16-18 UNF-3A	.5616	.5611	.877	.865	1.010	.328	.296	21/32	.078	.046	.835	.111	.070
AN180	5/8 -18 UNF-3A	.6240	.6234	.940	.928	1.090	.360	.328	47/64	.078	.046	.898	.111	.070
AN182	3/4 -16 UNF-3A	.7488	.7481	1.066	1.053	1.230	.422	.390	7/8	.078	.046	1.023	.111	.070
AN184	7/8 -14 UNF-3A	.8737	.8729	1.253	1.240	1.440	.485	.453	63/64	.094	.062	1.210	.111	.070
AN186	1 -14 NF-3A	.9985	.9975	1.441	1.428	1.660	.547	.515	1- 3/32	.094	.062	1.398	.111	.070

SEE SHEET 3 FOR NOTES (a), (b), AND (c).

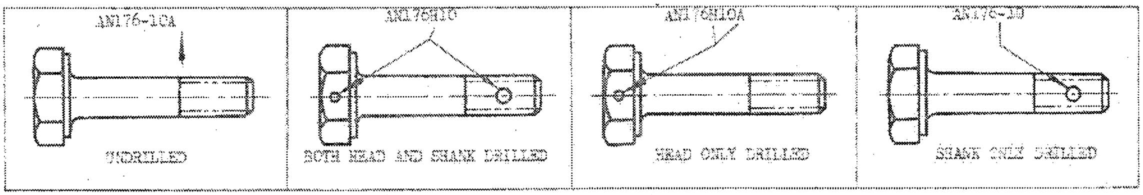
(d) THE DIAMETER OF THE WASHER FACE SHALL NOT EXCEED THE ACTUAL WIDTH ACROSS FLATS.

MATERIAL: NON-CORROSION-RESISTANT STEEL, CORROSION-RESISTANT STEEL, OR ALUMINUM ALLOY. SEE PROCUREMENT SPECIFICATION

FINISH: SEE PROCUREMENT SPECIFICATION, EXCEPT AS NOTED. PLATING MAY BE GROUND OFF SHANK OR UNDERSIDE OF HEAD. SMALL AREAS OF CADMIUM PLATING REMAINING ON THE GROUND PORTION OF BOLTS SHALL NOT BE CAUSE FOR REJECTION. UNPLATED SURFACES SHALL BE COATED WITH COMPOUND, SPECIFICATION MIL-C-11796 OR MIL-C-16173. SURFACE ROUGHNESS IN RMS MICRO INCHES BEFORE PLATING SHALL BE: 63 MAX FOR SHANK AND UNDERSIDE OF HEAD, 125 MAX FOR THREADS (VISUAL COMPARISON ONLY), 250 MAX FOR ALL OTHER AREAS (VISUAL COMPARISON ONLY).

ADD C BEFORE DASH NUMBER FOR CORROSION-RESISTANT STEEL BOLT. (SEE (c) NOTE.)
ADD DD BEFORE DASH NUMBER FOR ALUMINUM-ALLOY BOLT. (SEE (c) NOTE.)
ADD A AFTER DASH NUMBER FOR UNDRILLED BOLT. SEE ILLUSTRATION.
ADD H BEFORE DASH NUMBER FOR BOLT WITH DRILLED HEAD AND SHANK. SEE ILLUSTRATION.
ADD H BEFORE DASH NUMBER AND A AFTER DASH NUMBER FOR BOLT WITH DRILLED HEAD ONLY. SEE ILLUSTRATION

EXAMPLES OF PART NUMBERS:
AN176-10 = 3/8 NON-CORROSION-RESISTANT STEEL BOLT 1-5/64 LONG, 7/16 GRIP WITH DRILLED SHANK ONLY. SEE ILLUSTRATION.
AN176C10 = 3/8 CORROSION-RESISTANT STEEL BOLT 1-5/64 LONG, 7/16 GRIP WITH DRILLED SHANK ONLY. SEE ILLUSTRATION. (SEE (c) NOTE.)
AN176DD10 = 3/8 ALUMINUM-ALLOY BOLT 1-5/64 LONG, 7/16 GRIP WITH DRILLED SHANK ONLY. SEE ILLUSTRATION. (SEE (c) NOTE.)
AN176DD10A = 3/8 ALUMINUM-ALLOY BOLT 1-5/64 LONG, 7/16 GRIP, UNDRILLED SHANK AND HEAD. SEE ILLUSTRATION. (SEE (c) NOTE.)
AN176DDH10 = 3/8 ALUMINUM-ALLOY BOLT 1-5/64 LONG, 7/16 GRIP WITH DRILLED HEAD AND SHANK. SEE ILLUSTRATION. (SEE (c) NOTE.)
AN176DDH10A = 3/8 ALUMINUM-ALLOY BOLT 1-5/64 LONG, 7/16 GRIP WITH DRILLED HEAD ONLY. SEE ILLUSTRATION. (SEE (c) NOTE.)



BOLTS SHALL BE FREE FROM ALL HANGING BURRS AND SLIVERS WHICH MIGHT BECOME DISLODGED UNDER USAGE. COUNTERSINKING OF DRILLED HOLES IN HEAD IS MANDATORY. COUNTERSINKING OF DRILLED HOLES IN SHANK IS OPTIONAL.

DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: DECIMALS ±.010, ANGLES ±5°.

APPROVED 29 Mar 49 REVISED ① 14 Mar 52 ② 30 Oct 52 ③ 15 Mar 55 ④ 6 Apr 64

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PROCUREMENT SPECIFICATION
MIL-B-6812

AIR FORCE-NAVY AERONAUTICAL STANDARD

BOLT - MACHINE, CLOSE TOLERANCE, AIRCRAFT

AN173 THRU AN186

SUPERSEDES USAF DRAWINGS 83 THROUGH 96 AND NAS DRAWINGS 51 THROUGH 56.

DASH NO.	AN173		AN174		AN175		AN176		AN177		AN178		AN179	
	GRIP ±1/64	LENGTH +1/32 -1/64	GRIP ±1/64	LENGTH +1/32 -1/64	GRIP ±1/64	LENGTH +1/32 -1/64	GRIP ±1/64	LENGTH +1/32 -1/64	GRIP ±1/64	LENGTH +1/32 -1/64	GRIP ±1/64	LENGTH +1/32 -1/64	GRIP ±1/64	LENGTH +1/32 -1/64
3	1/16	15/32	1/16	15/32										
4	1/8	17/32	1/16	17/32	1/16	19/32								
5	1/4	21/32	3/16	21/32	3/16	23/32	1/16	45/64	1/16	23/32				
6	3/8	25/32	5/16	25/32	5/16	27/32	3/16	53/64	3/16	27/32	1/16	27/32	1/16	31/32
7	1/2	29/32	7/16	29/32	7/16	31/32	5/16	61/64	5/16	31/32	3/16	31/32	1/8	1- 1/32
10	5/8	1- 1/32	9/16	1- 1/32	9/16	1- 3/32	7/16	1- 5/64	7/16	1- 3/32	5/16	1- 3/32	1/4	1- 5/32
11	3/4	1- 5/32	11/16	1- 5/32	11/16	1- 7/32	9/16	1-13/64	9/16	1- 7/32	7/16	1- 7/32	3/8	1- 9/32
12	7/8	1- 9/32	13/16	1- 9/32	13/16	1-11/32	11/16	1-21/64	11/16	1-11/32	9/16	1-11/32	1/2	1-13/32
13	1	1-13/32	15/16	1-13/32	15/16	1-15/32	13/16	1-29/64	13/16	1-15/32	11/16	1-15/32	5/8	1-17/32
14	1-1/8	1-17/32	1- 1/16	1-17/32	1- 1/16	1-19/32	15/16	1-37/64	15/16	1-19/32	13/16	1-19/32	3/4	1-21/32
15	1-1/4	1-21/32	1- 3/16	1-21/32	1- 3/16	1-23/32	1- 1/16	1-45/64	1- 1/16	1-23/32	15/16	1-23/32	7/8	1-25/32
16	1-3/8	1-25/32	1- 5/16	1-25/32	1- 5/16	1-27/32	1- 3/16	1-53/64	1- 3/16	1-27/32	1- 1/16	1-27/32	1	1-29/32
17	1-1/2	1-29/32	1- 7/16	1-29/32	1- 7/16	1-31/32	1- 5/16	1-61/64	1- 5/16	1-31/32	1- 3/16	1-31/32	1-1/8	2- 1/32
20	1-5/8	2- 1/32	1- 9/16	2- 1/32	1- 9/16	2- 3/32	1- 7/16	2- 5/64	1- 7/16	2- 3/32	1- 5/16	2- 3/32	1-1/4	2- 5/32
21	1-3/4	2- 5/32	1-11/16	2- 5/32	1-11/16	2- 7/32	1- 9/16	2-13/64	1- 9/16	2- 7/32	1- 7/16	2- 7/32	1-3/8	2- 9/32
22	1-7/8	2- 9/32	1-13/16	2- 9/32	1-13/16	2-11/32	1-11/16	2-21/64	1-11/16	2-11/32	1- 9/16	2-11/32	1-1/2	2-13/32
23	2	2-13/32	1-15/16	2-13/32	1-15/16	2-15/32	1-13/16	2-29/64	1-13/16	2-15/32	1-11/16	2-15/32	1-5/8	2-17/32
24	2-1/8	2-17/32	2- 1/16	2-17/32	2- 1/16	2-19/32	1-15/16	2-37/64	1-15/16	2-19/32	1-13/16	2-19/32	1-3/4	2-21/32
25	2-1/4	2-21/32	2- 3/16	2-21/32	2- 3/16	2-23/32	2- 1/16	2-45/64	2- 1/16	2-23/32	1-15/16	2-23/32	1-7/8	2-25/32
26	2-3/8	2-25/32	2- 5/16	2-25/32	2- 5/16	2-27/32	2- 3/16	2-53/64	2- 3/16	2-27/32	2- 1/16	2-27/32	2	2-29/32
27	2-1/2	2-29/32	2- 7/16	2-29/32	2- 7/16	2-31/32	2- 5/16	2-61/64	2- 5/16	2-31/32	2- 3/16	2-31/32	2-1/8	3- 1/32
30	2-5/8	3- 1/32	2- 9/16	3- 1/32	2- 9/16	3- 3/32	2- 7/16	3- 5/64	2- 7/16	3- 3/32	2- 5/16	3- 3/32	2-1/4	3- 5/32
31	2-3/4	3- 5/32	2-11/16	3- 5/32	2-11/16	3- 7/32	2- 9/16	3-13/64	2- 9/16	3- 7/32	2- 7/16	3- 7/32	2-3/8	3- 9/32
32	2-7/8	3- 9/32	2-13/16	3- 9/32	2-13/16	3-11/32	2-11/16	3-21/64	2-11/16	3-11/32	2- 9/16	3-11/32	2-1/2	3-13/32
33	3	3-13/32	2-15/16	3-13/32	2-15/16	3-15/32	2-13/16	3-29/64	2-13/16	3-15/32	2-11/16	3-15/32	2-5/8	3-17/32
34	3-1/8	3-17/32	3- 1/16	3-17/32	3- 1/16	3-19/32	2-15/16	3-37/64	2-15/16	3-19/32	2-13/16	3-19/32	2-3/4	3-21/32
35	3-1/4	3-21/32	3- 3/16	3-21/32	3- 3/16	3-23/32	3- 1/16	3-45/64	3- 1/16	3-23/32	2-15/16	3-23/32	2-7/8	3-25/32
36	3-3/8	3-25/32	3- 5/16	3-25/32	3- 5/16	3-27/32	3- 3/16	3-53/64	3- 3/16	3-27/32	3- 1/16	3-27/32	3	3-29/32
37	3-1/2	3-29/32	3- 7/16	3-29/32	3- 7/16	3-31/32	3- 5/16	3-61/64	3- 5/16	3-31/32	3- 3/16	3-31/32	3-1/8	4- 1/32
40	3-5/8	4- 1/32	3- 9/16	4- 1/32	3- 9/16	4- 3/32	3- 7/16	4- 5/64	3- 7/16	4- 3/32	3- 5/16	4- 3/32	3-1/4	4- 5/32
41	3-3/4	4- 5/32	3-11/16	4- 5/32	3-11/16	4- 7/32	3- 9/16	4-13/64	3- 9/16	4- 7/32	3- 7/16	4- 7/32	3-3/8	4- 9/32
42	3-7/8	4- 9/32	3-13/16	4- 9/32	3-13/16	4-11/32	3-11/16	4-21/64	3-11/16	4-11/32	3- 9/16	4-11/32	3-1/2	4-13/32
43	4	4-13/32	3-15/16	4-13/32	3-15/16	4-15/32	3-13/16	4-29/64	3-13/16	4-15/32	3-11/16	4-15/32	3-5/8	4-17/32
44	4-1/8	4-17/32	4- 1/16	4-17/32	4- 1/16	4-19/32	3-15/16	4-37/64	3-15/16	4-19/32	3-13/16	4-19/32	3-3/4	4-21/32
45	4-1/4	4-21/32	4- 3/16	4-21/32	4- 3/16	4-23/32	4- 1/16	4-45/64	4- 1/16	4-23/32	3-15/16	4-23/32	3-7/8	4-25/32
46	4-3/8	4-25/32	4- 5/16	4-25/32	4- 5/16	4-27/32	4- 3/16	4-53/64	4- 3/16	4-27/32	4- 1/16	4-27/32	4	4-29/32
47	4-1/2	4-29/32	4- 7/16	4-29/32	4- 7/16	4-31/32	4- 5/16	4-61/64	4- 5/16	4-31/32	4- 3/16	4-31/32	4-1/8	5- 1/32
50	4-5/8	5- 1/32	4- 9/16	5- 1/32	4- 9/16	5- 3/32	4- 7/16	5- 5/64	4- 7/16	5- 3/32	4- 5/16	5- 3/32	4-1/4	5- 5/32
51	4-3/4	5- 5/32	4-11/16	5- 5/32	4-11/16	5- 7/32	4- 9/16	5-13/64	4- 9/16	5- 7/32	4- 7/16	5- 7/32	4-3/8	5- 9/32
52	4-7/8	5- 9/32	4-13/16	5- 9/32	4-13/16	5-11/32	4-11/16	5-21/64	4-11/16	5-11/32	4- 9/16	5-11/32	4-1/2	5-13/32
53	5	5-13/32	4-15/16	5-13/32	4-15/16	5-15/32	4-13/16	5-29/64	4-13/16	5-15/32	4-11/16	5-15/32	4-5/8	5-17/32
54	5-1/8	5-17/32	5- 1/16	5-17/32	5- 1/16	5-19/32	4-15/16	5-37/64	4-15/16	5-19/32	4-13/16	5-19/32	4-3/4	5-21/32
55	5-1/4	5-21/32	5- 3/16	5-21/32	5- 3/16	5-23/32	5- 1/16	5-45/64	5- 1/16	5-23/32	4-15/16	5-23/32	4-7/8	5-25/32
56	5-3/8	5-25/32	5- 5/16	5-25/32	5- 5/16	5-27/32	5- 3/16	5-53/64	5- 3/16	5-27/32	5- 1/16	5-27/32	5	5-29/32
57	5-1/2	5-29/32	5- 7/16	5-29/32	5- 7/16	5-31/32	5- 5/16	5-61/64	5- 5/16	5-31/32	5- 3/16	5-31/32	5-1/8	6- 1/32
60	5-5/8	6- 1/32	5- 9/16	6- 1/32	5- 9/16	6- 3/32	5- 7/16	6- 5/64	5- 7/16	6- 3/32	5- 5/16	6- 3/32	5-1/4	6- 5/32
61	5-3/4	6- 5/32	5-11/16	6- 5/32	5-11/16	6- 7/32	5- 9/16	6-13/64	5- 9/16	6- 7/32	5- 7/16	6- 7/32	5-3/8	6- 9/32
62	5-7/8	6- 9/32	5-13/16	6- 9/32	5-13/16	6-11/32	5-11/16	6-21/64	5-11/16	6-11/32	5- 9/16	6-11/32	5-1/2	6-13/32
63	6	6-13/32	5-15/16	6-13/32	5-15/16	6-15/32	5-13/16	6-29/64	5-13/16	6-15/32	5-11/16	6-15/32	5-5/8	6-17/32
64	6-1/8	6-17/32	6- 1/16	6-17/32	6- 1/16	6-19/32	5-15/16	6-37/64	5-15/16	6-19/32	5-13/16	6-19/32	5-3/4	6-21/32
65	6-1/4	6-21/32	6- 3/16	6-21/32	6- 3/16	6-23/32	6- 1/16	6-45/64	6- 1/16	6-23/32	5-15/16	6-23/32	5-7/8	6-25/32
66	6-3/8	6-25/32	6- 5/16	6-25/32	6- 5/16	6-27/32	6- 3/16	6-53/64	6- 3/16	6-27/32	6- 1/16	6-27/32	6	6-29/32
67	6-1/2	6-29/32	6- 7/16	6-29/32	6- 7/16	6-31/32	6- 5/16	6-61/64	6- 5/16	6-31/32	6- 3/16	6-31/32	6-1/8	7- 1/32
70	6-5/8	7- 1/32	6- 9/16	7- 1/32	6- 9/16	7- 3/32	6- 7/16	7- 5/64	6- 7/16	7- 3/32	6- 5/16	7- 3/32	6-1/4	7- 5/32
71	6-3/4	7- 5/32	6-11/16	7- 5/32	6-11/16	7- 7/32	6- 9/16	7-13/64	6- 9/16	7- 7/32	6- 7/16	7- 7/32	6-3/8	7- 9/32
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73	7	7-13/32	6-15/16	7-13/32	6-15/16	7-15/32	6-13/16	7-29/64	6-13/16	7-15/32	6-11/16	7-15/32	6-5/8	7-17/32
74	7-1/8	7-17/32	7- 1/16	7-17/32	7- 1/16	7-19/32	6-15/16	7-37/64	6-15/16	7-19/32	6-13/16	7-19/32	6-3/4	7-21/32
75	7-1/4	7-21/32	7- 3/16	7-21/32	7- 3/16	7-23/32	7- 1/16	7-45/64	7- 1/16	7-23/32	6-15/16	7-23/32	6-7/8	7-25/32
76	7-3/8	7-25/32	7- 5/16	7-25/32	7- 5/16	7-27/32	7- 3/16	7-53/64	7- 3/16	7-27/32	7- 1/16	7-27/32	7	7-29/32
77	7-1/2	7-29/32	7- 7/16	7-29/32	7- 7/16	7-31/32	7- 5/16	7-61/64	7- 5/16	7-31/32	7- 3/16	7-31/32	7-1/8	8- 1/32
80	7-5/8	8- 1/32	7- 9/16	8- 1/32	7- 9/16	8- 3/32	7- 7/16	8- 5/64	7- 7/16	8- 3/32	7- 5/16	8- 3/32	7-1/4	8- 5/32
81					7-11/16	8- 7/32					7- 7/16	8- 7/32		
82											7- 9/16	8-11/32		
83											7-11/16	8-15/32		
84											7-13/16	8-19/32		
85							8- 1/16	8-45/64			7-15/16	8-23/32		
86											8- 1/16	8-27/32		
87											8- 3/16	8-31/32		

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APPROVED 29 Mar 49 REVISED ① 24 Mar 52 ② FOR CHANGES SEE SHEET 1

PROCUREMENT SPECIFICATION MIL-B-6812	AIR FORCE-NAVY AERONAUTICAL STANDARD BOLT - MACHINE, CLOSE TOLERANCE, AIRCRAFT	AN173 THRU AN186 SHEET 2 OF 4

NOTE: This drawing was approved by joint action of the [redacted] AIR FORCE and Navy Departments as the [redacted] AIR FORCE-Navy standard for this product. This drawing supersedes all antecedent standard drawings for the same product and shall become effective for the procurement of aeronautical supplies, or for use in new design, not later than 6 months after the latest date of approval shown. It may be put into effect, however, at an earlier date after promulgation.

N[illegible]E: When Government drawings, specifications, or other data are used for any purpose other than in connection with a defin[illegible] related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

DASH NO.	AN180 GRIP ±1/64	AN180 LENGTH +1/32 -1/64	AN182 GRIP ±1/64	AN182 LENGTH +1/32 -1/64	AN184 GRIP ±1/64	AN184 LENGTH +1/32 -1/64	AN186 GRIP ±1/64	AN186 LENGTH +1/32 -1/64
7	1/16	1- 1/64						
10	3/16	1- 9/64	1/16	1- 5/32				
11	5/16	1-17/64	3/16	1- 9/32	1/16	1- 5/16		
12	7/16	1-25/64	5/16	1-13/32	3/16	1- 7/16	1/8	1-1/2
13	9/16	1-33/64	7/16	1-17/32	5/16	1- 9/16	1/4	1-5/8
14	11/16	1-41/64	9/16	1-21/32	7/16	1-11/16	3/8	1-3/4
15	13/16	1-49/64	11/16	1-25/32	9/16	1-13/16	1/2	1-7/8
16	15/16	1-57/64	13/16	1-29/32	11/16	1-15/16	5/8	2
17	1- 1/16	2- 1/64	15/16	2- 1/32	13/16	2- 1/16	3/4	2-1/8
20	1- 3/16	2- 9/64	1- 1/16	2- 5/32	15/16	2- 3/16	7/8	2-1/4
21	1- 5/16	2-17/64	1- 3/16	2- 9/32	1- 1/16	2- 5/16	1	2-3/8
22	1- 7/16	2-25/64	1- 5/16	2-13/32	1- 3/16	2- 7/16	1-1/8	2-1/2
23	1- 9/16	2-33/64	1- 7/16	2-17/32	1- 5/16	2- 9/16	1-1/4	2-5/8
24	1-11/16	2-41/64	1- 9/16	2-21/32	1- 7/16	2-11/16	1-3/8	2-3/4
25	1-13/16	2-49/64	1-11/16	2-25/32	1- 9/16	2-13/16	1-1/2	2-7/8
26	1-15/16	2-57/64	1-13/16	2-29/32	1-11/16	2-15/16	1-5/8	3
27	2- 1/16	3- 1/64	1-15/16	3- 1/32	1-13/16	3- 1/16	1-3/4	3-1/8
30	2- 3/16	3- 9/64	2- 1/16	3- 5/32	1-15/16	3- 3/16	1-7/8	3-1/4
31	2- 5/16	3-17/64	2- 3/16	3- 9/32	2- 1/16	3- 5/16	2	3-3/8
32	2- 7/16	3-25/64	2- 5/16	3-13/32	2- 3/16	3- 7/16	2-1/8	3-1/2
33	2- 9/16	3-33/64	2- 7/16	3-17/32	2- 5/16	3- 9/16	2-1/4	3-5/8
34	2-11/16	3-41/64	2- 9/16	3-21/32	2- 7/16	3-11/16	2-3/8	3-3/4
35	2-13/16	3-49/64	2-11/16	3-25/32	2- 9/16	3-13/16	2-1/2	3-7/8
36	2-15/16	3-57/64	2-13/16	3-29/32	2-11/16	3-15/16	2-5/8	4
37	3- 1/16	4- 1/64	2-15/16	4- 1/32	2-13/16	4- 1/16	2-3/4	4-1/8
40	3- 3/16	4- 9/64	3- 1/16	4- 5/32	2-15/16	4- 3/16	2-7/8	4-1/4
41	3- 5/16	4-17/64	3- 3/16	4- 9/32	3- 1/16	4- 5/16	3	4-3/8
42	3- 7/16	4-25/64	3- 5/16	4-13/32	3- 3/16	4- 7/16	3-1/8	4-1/2
43	3- 9/16	4-33/64	3- 7/16	4-17/32	3- 5/16	4- 9/16	3-1/4	4-5/8
44	3-11/16	4-41/64	3- 9/16	4-21/32	3- 7/16	4-11/16	3-3/8	4-3/4
45	3-13/16	4-49/64	3-11/16	4-25/32	3- 9/16	4-13/16	3-1/2	4-7/8
46	3-15/16	4-57/64	3-13/16	4-29/32	3-11/16	4-15/16	3-5/8	5
47	4- 1/16	5- 1/64	3-15/16	5- 1/32	3-13/16	5- 1/16	3-3/4	5-1/8
50	4- 3/16	5- 9/64	4- 1/16	5- 5/32	3-15/16	5- 3/16	3-7/8	5-1/4
51	4- 5/16	5-17/64	4- 3/16	5- 9/32	4- 1/16	5- 5/16	4	5-3/8
52	4- 7/16	5-25/64	4- 5/16	5-13/32	4- 3/16	5- 7/16	4-1/8	5-1/2
53	4- 9/16	5-33/64	4- 7/16	5-17/32	4- 5/16	5- 9/16	4-1/4	5-5/8
54	4-11/16	5-41/64	4- 9/16	5-21/32	4- 7/16	5-11/16	4-3/8	5-3/4
55	4-13/16	5-49/64	4-11/16	5-25/32	4- 9/16	5-13/16	4-1/2	5-7/8
56	4-15/16	5-57/64	4-13/16	5-29/32	4-11/16	5-15/16	4-5/8	6
57	5- 1/16	6- 1/64	4-15/16	6- 1/32	4-13/16	6- 1/16	4-3/4	6-1/8
60	5- 3/16	6- 9/64	5- 1/16	6- 5/32	4-15/16	6- 3/16	4-7/8	6-1/4
61	5- 5/16	6-17/64	5- 3/16	6- 9/32	5- 1/16	6- 5/16	5	6-3/8
62	5- 7/16	6-25/64	5- 5/16	6-13/32	5- 3/16	6- 7/16	5-1/8	6-1/2
63	5- 9/16	6-33/64	5- 7/16	6-17/32	5- 5/16	6- 9/16	5-1/4	6-5/8
64	5-11/16	6-41/64	5- 9/16	6-21/32	5- 7/16	6-11/16	5-3/8	6-3/4
65	5-13/16	6-49/64	5-11/16	6-25/32	5- 9/16	6-13/16	5-1/2	6-7/8
66	5-15/16	6-57/64	5-13/16	6-29/32	5-11/16	6-15/16	5-5/8	7
67	6- 1/16	7- 1/64	5-15/16	7- 1/32	5-13/16	7- 1/16	5-3/4	7-1/8
70	6- 3/16	7- 9/64	6- 1/16	7- 5/32	5-15/16	7- 3/16	5-7/8	7-1/4
71	6- 5/16	7-17/64	6- 3/16	7- 9/32	6- 1/16	7- 5/16	6	7-3/8
72	6- 7/16	7-25/64	6- 5/16	7-13/32	6- 3/16	7- 7/16	6-1/8	7-1/2
73	6- 9/16	7-33/64	6- 7/16	7-17/32	6- 5/16	7- 9/16	6-1/4	7-5/8
74	6-11/16	7-41/64	6- 9/16	7-21/32	6- 7/16	7-11/16	6-3/8	7-3/4
75	6-13/16	7-49/64	6-11/16	7-25/32	6- 9/16	7-13/16	6-1/2	7-7/8
76	6-15/16	7-57/64	6-13/16	7-29/32	6-11/16	7-15/16	6-5/8	8
77	7- 1/16	8- 1/64	6-15/16	8- 1/32	6-13/16	8- 1/16	6-3/4	8-1/8
80	7- 3/16	8- 9/64	7- 1/16	8- 5/32	6-15/16	8- 3/16	6-7/8	8-1/4
81							7	8-3/8
82							7-1/8	8-1/2

(a) MARKING MAY BE EITHER RAISED OR DEPRESSED.
THE HEIGHT OR DEPTH OF THE MARKINGS SHALL NOT EXCEED THE FOLLOWING VALUES:
.025 INCH FOR 9/16 DIAMETER BOLTS AND SMALLER;
.030 INCH FOR 5/8 DIAMETER BOLTS AND LARGER.
HEAD MARKING MAY INCLUDE MANUFACTURER'S MARKINGS.

(b) GRIP LENGTH OF BOLTS SHALL BE MEASURED FROM THE UNDERSIDE OF THE HEAD TO THE END OF THE FULL CYLINDRICAL PORTION OF THE SHANK. COMPLETE THREADS SHALL BEGIN WITHIN TWO THREAD PITCH MAXIMUM. TWO THREAD PITCH MAXIMUM MAY CONSIST OF INCOMPLETE THREAD OR EXTRUSION ANGLE.

(c) CORROSION RESISTING STEEL BOLTS AND ALUMINUM ALLOY BOLTS INACTIVE FOR DESIGN AFTER 24 MARCH 1952.

FOR DEFINITION AND APPLICATION OF DRAWING STATUS NOTES, SEE ANA BULLETIN NO. 337.

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PROCUREMENT SPECIFICATION
MIL-B-6812

AIR FORCE-NAVY AERONAUTICAL STANDARD

BOLT - MACHINE, CLOSE TOLERANCE, AIRCRAFT

AN173 THRU AN186

BASIC AN PART NO.	THREAD T	RATED STRENGTH (POUNDS)					
		ULTIMATE TENSILE AT ROOT DIA		YIELD TENSILE AT ROOT DIA		SINGLE SHEAR AT FULL DIA	
		STEEL	AL ALLOY	STEEL	AL ALLOY	STEEL	AL ALLOY
AN173	NO. 10-32 NF-3A	2 210	1 100	1 690	710	2 125	990
AN174	1/4 -28 UNF-3A	4 060	2 030	3 130	1 310	3 680	1 715
AN175	5/16-24 UNF-3A	6 500	3 220	4 980	2 080	5 750	2 685
AN176	3/8 -24 UNF-3A	10 100	5 020	7 740	3 240	8 280	3 870
AN177	7/16-20 UNF-3A	13 600	6 750	10 430	4 350	11 250	5 250
AN178	1/2 -20 UNF-3A	18 500	9 180	14 190	5 920	14 700	6 850
AN179	9/16-18 UNF-3A	23 600	11 700	18 100	7 550	18 700	8 700
AN180	5/8 -18 UNF-3A	30 100	14 900	23 080	9 610	23 000	10 750
AN182	3/4 -16 UNF-3A	44 000	21 800	33 730	14 100	33 150	15 500
AN184	7/8 -14 UNF-3A	60 000	29 800	46 000	19 200	45 050	21 050
AN186	1 -14 NF-3A	80 700	40 000	61 870	25 800	58 900	27 500

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AN173 THRU AN186